

EDUCATION

• MSc Computer Science [Grade: 1.6]

University of Stuttgart

Apr 2024 - ongoing

Stuttgart, Germany

- Specialisation in Autonomous Systems with course work in the domains of Machine Learning, Deep Learning, GenAI, Information Visualisation, Computer Vision, Reinforcement Learning among others.
- Completed my master thesis titled **Proof-Guided Neuro-Symbolic Reasoning for Large Language Models**.
- Built an end-to-end continual learning pipeline called **LiverLLM** finetuned on curated medical data to support understanding of liver diseases and liver cancer.
- Seminar paper on **LLM based Planning** discussing different ways LLMs can be used in plan generation and evaluation.
- Awarded the **Femtec Scholarship** and inducted into Femtec Class 46.

• BSc Computer Science [Grade: 1.9]

St. Xavier's College

Jul 2018 - Jun 2021

Kolkata, India

- Bachelor Thesis: Developed a new **cryptographic algorithm** to encrypt small texts, passwords and OTPs focusing on speed and security. Programming language used: **Java**.
- Awarded the Fr. Lafont Scholarship for Academic Performance.

TECHNICAL SKILLS

- **AI / Machine Learning**: Large Language Models (LLMs), **SFT**, **GRPO**, **RLHF**, **RLVR**, Knowledge Graphs, Ontologies, RAG, Agentic AI, Multi-Agent Systems, Continual Learning, Prompt Engineering, LangChain, Vector Databases, **PyTorch**, **Scikit-learn**
- **Languages & Frameworks**: Python, Flask, Angular, ReactJS, ReactNative, Streamlit, HTML/CSS, JavaScript, REST API, Bootstrap, SQL, MongoDB, SPARQL, RDF, OWL
- **Tools**: Git, Linux, NumPy, Pandas, RDFox

EXPERIENCE

• Mercedes-Benz Technology Innovation

Work Student

Jun 2025 – Sept 2025

Stuttgart, Germany

- Developed frontend features for quality assurance (QA) of navigation systems (using **Angular**, **Golang** and **MongoDB**)
- Implemented backend logic for new metrics for QA (using **Golang** and **MongoDB**)
- Implemented **Pydantic** test cases to evaluate LLMs on structured outputs for single and multi-turn conversations

• Daimler Trucks AG

Work Student, **PMO Support** standardizing the on/off-boarding pipeline.

Sept 2024 – Feb 2025

Stuttgart, Germany

- Implemented standardization of the on/off-boarding pipeline by creating a central tracking dashboard for the internal stakeholder usage.

• Fraunhofer IAO

Student Research Assistant

Jul 2024 – Mar 2025

Stuttgart, Germany

- Implemented a proof of concept using **SpringBoot**, **Apache Kafka**, and **Docker** to demonstrate secure communication.
- Led research and evaluation of optimal **text-to-speech** implementations for **ReactNative** applications.

• Spring and River

Software Engineer

Feb 2023 – Mar 2024

Kolkata, India

- Developed the company's flagship product using **Angular**, **Python Flask**, **REST API** and **MaterialUI**
- Led the development of an **e-commerce** application using **ReactJS** and the companion mobile app using **ReactNative**.

• PwC India

Technology Consultant

Aug 2021 – Feb 2023

Bengaluru, India

- * Implemented custom solutions for tax, accounting, sales and other modules.
- * Optimized creation of custom sales reports, invoices and billing information to reduce manual effort by 80%.

PROJECTS

• Proof-Guided Neuro-Symbolic Reasoning for Large Language Models

Dec 2025 - Sept 2026

LLMs, Symbolic Reasoning, OWL/RDF, SFT, RLHF

- Developing a pipeline to improve symbolic reasoning in LLMs using **OWL-2-RL** rules.
- Implemented QA baselines with binary QAs and MCQs, by introducing distractors, to evaluate model reasoning.
- Developed a two-stage training process with Supervised Fine-tuning (**SFT**) and Reinforcement Learning with Verifiable Rewards (**RLVR**) to safely train and update model parameters to improve reasoning correctness and faithfulness.
- Evaluated models on their direct entailment generation capabilities with intermediary proof traces.

• STARTARCH – Agentic AI Research Commercialization Platform

Jun 2026

[Hackathon Project - GitHub](#)

Agentic AI, Gemini, Pioneer, Tavily, Flask, React

- Built a multi-agent AI platform that converts academic research papers into commercialization opportunities, giving detailed market assessments, and investor-style venture recommendations.
- Fine-tuned a Pioneer GLiNER2 model using synthetic data generation to classify commercialization bottlenecks across market dimensions.
- Leveraged Gemini and Tavily to perform automated competitor analysis, market sizing, funding research, regulatory assessment, and partnership discovery.
- Reduced workflow execution latency through parallel agent execution and concurrent processing of independent opportunity analysis tracks.

• LiverLLM

[GitHub](#)

Apr 2025 - Jul 2025

LLMs, Continual Learning, Fine-Tuning, RAG

- Built an end-to-end continual learning pipeline to fine-tune a Llama-3-8B model on domain-specific medical data.
- Curated datasets from liver cancer research publications and targeted QA pairs for liver diseases.
- Addressed static knowledge limitations in LLMs via domain adaptation and retrieval-augmented generation (RAG).

• AI Job Application Assistant

[GitHub](#)

Apr 2025 - Jul 2025

LLMs, Groq, Web-Scraping, Flask, ReactJS

- Built an end-to-end web-app to support and streamline the job application process.
- Used LLMs to process job content and user documents (CVs, cover letters), and then customize the documents to align with the job description.

• Kanban Board

[Frontend](#) / [Backend](#)

Jan 2025 - Feb 2025

Web App, Angular, Flask

- Developed a full-stack Kanban web application using Angular 19 and Flask.
- Integrated MongoDB for user management and Kanban state persistence.
- Implemented REST APIs for authentication and CRUD operations.

• Monolithic Cloud Architecture

[Project Report](#)

Feb 2025 - Feb 2025

Research Project, AWS

- Designed a low-cost, cloud-native system architecture on AWS for a growing e-commerce business.
- Implemented an AWS hosting stack using EC2, RDS (MySQL), S3, CloudFront, and Route 53 with auto-scaling and CDN support.
- Integrated WordPress and WooCommerce, justifying trade-offs in consistency, security, and scalability.

CERTIFICATIONS

• 5-Day Gen AI Intensive

[Badge](#)

Mar 2025 - Mar 2025

LLMs, RAG, Gemini

ACHIEVEMENTS

• Femtec Scholarship Holder

Sept 2024 - Sept 2025

- One of the scholarship holders among 55 others across all TU9 Germany
- Femtec is an international scholarship program directed towards **women in STEM**, guiding them in career building, leadership and networking.
- Member of class 46.

• Femtec Innovation Lab with E.ON GmbH

Mar 2025 - Jul 2025

- Worked on a collaborative research project, investigating sustainability and affordability aspects in energy systems.

• Awarded the Online Stipendium from e-fellows.net